Team: Batties

Client: Cornell CALS Extension, E&J Gallo Winery, and National Grape.

Problem Statement: Spotted Lantern Flies (SLF) are a major concern to the grape industry because up to 400 SLF can feed on each vine in a vineyard (Cornell CALS, n.d.). Once SLF reach the vineyard, they are very difficult to exterminate, as they're in a dense environment, have unpredictable movement, and a relatively fast reaction time. Over time, the repetitive feeding leads to poor harvests, which can cause up to 90% yield loss (Pfeiffer et al, 2022).

Why it matters to the end user: It is essential that the grapes survive to harvest time, which becomes a challenge as long as SLF continue to feed on the vines. If we attempt to eradicate the SLF before they reach the vineyard, they will be less likely to contaminate the juice during harvest. Narrowing our focus to defending vineyards allows us to create an industry specific solution in that we can ensure that our device is more attractive than grapes specifically. This narrow scope was an important reason we chose this problem over attempting to target SLF eggs, as they can be found almost anywhere at any height.

Proposed Direction:

- **Flytrap Proposal:** We plan to use an attract and capture method where SLF are lured by a scent and then trapped in one of three capture mechanisms we are considering:

1. Sticky surface capture – insects contact an adhesive surface after entering and are immobilized.
2. Low-power fan containment – fan pulls insects into a collection chamber and prevents escape.
3. Soap-water capture – insects fall into a liquid reservoir (dish soap and water) that prevents escape

A perimeter can be made around vineyards, or these traps can be hung from trees. We can test traps around the Tree of Heaven (the location the SLF will most likely be at this time) to determine the necessary radius and amount of traps to be used. The long-term product would consist of many low-cost, weather-resistant units placed strategically throughout a vineyard to reduce pest populations without chemical spraying and with minimal maintenance.

Our prototype will be a single working trap that demonstrates the feasibility of the attract-and-capture mechanism. The semester goal is not full vineyard deployment, but validation of the core mechanical principles. We will build prototypes for one-way entrance chambers to capture the SLF. This proof-of-concept will establish whether SLF can be reliably attracted and captured using passive luring plus mechanical containment, informing later design iterations, scaling strategy, and field placement density.

Risks and Unknowns: The greatest risk would come from focusing solely on deterring the SLF from the vineyards rather than actually killing them. The patterns of the SLF are fairly unpredictable (according to experts, they feed on over 100 different American plants), so if we only keep them from feeding on grapes, they may choose a different crop to feed on, which could just move the problem to a different industry. If we choose to use heavy machinery or pesticides, we could also run the risk of destroying the grapes through our proposed solution. While the experts say that the grapes do not rely on pollinating insects, we should still be careful not to hurt any lifeforms that are beneficial to the ecosystem. One unknown we have is the attractiveness of different chemicals or food products to SLF.

Questions

- Since we want something to attract the lantern flies, we would like to know if

Question 1: Have there been substances that have been found to be more attractive to SLF than grape vines, and are they attracted to the scent, the sight, or something else?
Decision affected: We need to know what the most effective attraction mechanism is for our flytrap.

Question 2: Which other insects are attracted to the same items as SLF?
Decision affected: We want to attempt to minimize risk to the native species.

Question 3: What is the size of the SLF at the time they usually make their way into the vineyards?
Decision affected: We would like to ensure that our nets are the right size.

References

Pfeiffer, D.G. et al. (2022) *Spotted lanternfly in virginia vineyards: Lycorma Delicatula (white) (Hemiptera: Fulgoridae)*. doi:10.21061/ento-323np.

Spotted lanternfly damage. Cornell CALS. (n.d.). https://cals.cornell.edu/integrated-pest-management/outreach-education/whats-bugging-you/spotted-lanternfly/spotted-lanternfly-damage


```
# Your Project Title
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Spotted Lantern Flytrap

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**Team:** Batties
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**Client(s):** Cornell CALS Extension / E\&J Gallo Winery / National Grape
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Client: Cornell CALS Extension, E&J Gallo Winery, and National Grape.

```
## Problem statement (most important)
```

Problem Statement

Up to 400 SLF can feed on each vine in a vineyard (Cornell CALS, n.d.) Repetitive feeding can cause up to 90% yield loss (Pfeiffer et al, 2022).

```
## Impact
Give context for why solving this sub-problem addresses the client's user
needs.
```

Impact

As long as SLF continue to feed on grape vines, the grapes will be less likely to survive to harvest. Narrowing our focus to defending vineyards allows us to create an industry-specific solution.

```
## Proposed direction(s)
### Concept A (primary): <1-line name>
```

Concept: Spotted Lantern Flytrap

```
**What it is:** <1-2 sentences.>
```

Attract and capture: We are exploring sticky surface capture (insects stick to an adhesive), low-power fan containment (a fan pulls insects into a collection chamber), or soap-water capture (use of a liquid reservoir).

```
**How it would be used:** <2-4 bullet "user flow" steps.>
```

- Users hang traps on trees
- Users check and clear out traps periodically

```
**Why it's better than the status quo:** <2-3 bullets.>
```

- There are too many SLF to try and kill by hand.
- There will be less chances of contamination later on.

```
**End-of-semester proof-of-concept:** <what you can realistically build/test in MAE 2250.>
```

A single one-way entrance chamber with bait inside will test whether SLF can be reliably attracted and captured using passive bait plus mechanical containment.

```
## Key risks / unknowns

- <Risk 1> — why it matters, and how you'll test it.
```

Risk 1: Capturing pollinators: Many are already endangered. We will test our bait outside to see what is attracted to it.

```
- <Risk 2> — why it matters, and how you'll test it.
```

Risk 2: SLF ignoring the bait: If they prefer the grapes, our traps will not work. We will place various baits (including grape vines) in front of SLF and seeing which they choose.

```
- <Risk 3> — why it matters, and how you'll test it.
```

Risk 3: SLF escaping the trap: This will negate our goal.. We will add SLF and SLF-sized objects to the prototype to test this possibility.

```
## Questions for the client
Focus on questions they can answer from lived experience.
1. **<Question 1>**
   *Decision affected:* <what choice this answer changes for your team.>
2. **<Question 2>**
```

```
   *Decision affected:* <what choice this answer changes for your team.>
3. **<Question 3>**
   *Decision affected:* <what choice this answer changes for your team.>
4. **<Question 4 (optional)>**
```

Question 1: Are there substances that SLF like more than grapes, and are they attracted to the scent, the sight, or something else?
Decision affected: We need to know what is the most effective bait.

Question 2: Which other insects are attracted to the same substances as SLF?
Decision affected: This will also inform bait used.

Question 3: What is the size and weight of the SLF at the time they enter the vineyards?
Decision affected: We would like to choose the optimal net size.

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   *Decision affected:* <what choice this answer changes for your team.>

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# Optional page 2 — References and/or one figure

## References

- <Source 1 (URL or citation)>
- <Source 2 (URL or citation)>
```

References

Pfeiffer, D.G. et al. (2022) *Spotted lanternfly in virginia vineyards: Lycorma Delicatula (white) (Hemiptera: Fulgoridae)*. doi:10.21061/ento-323np.

Spotted lanternfly damage. Cornell CALS. (n.d.). https://cals.cornell.edu/integrated-pest-management/outreach-education/whats-bugging-you/spotted-lanternfly/spotted-lanternfly-damage


```
## Figure
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Spotted Lantern Flytrap

Team: Batties

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Problem Statement

Up to 400 SLF can feed on each vine in a vineyard (Cornell CALS, n.d.) Over time, the repetitive feeding can cause up to 90% yield loss (Pfeiffer et al, 2022).

Impact

As long as SLF continue to feed on grape vines, the grapes will be less likely to survive to harvest. If we attempt to eradicate the SLF before they reach the vineyard, they will be less likely to contaminate the juice as well. Narrowing our focus to defending vineyards allows us to create an industry-specific solution.

Proposed Direction

Concept: Spotted Lantern Flytrap

What it is: We plan to use an attract and capture method where SLF are lured by a scent and then trapped in a container. We are exploring sticky surface capture (contact to an adhesive), low-power fan containment (a fan pulling insects into a collection chamber), or soap-water capture (use of a liquid reservoir).

How it will be used:

- Hang traps on trees
- Check and clear out traps periodically

Why it's better than the status quo:

- There are too many SLF to try and kill by hand.
- There will be less chances of contamination later on.

End-of-semester proof-of-concept:

We will prototype one-way entrance chambers to capture the SLF with a lure inside. This proof-of-concept will establish whether SLF can be reliably attracted and captured using passive bait plus mechanical containment.

Questions for the client:

Have there been substances that have been found to be more attractive to SLF than grape vines, and are they attracted to the scent, the sight, or something else?

Decision affected: We need to know what is the most effective bait.

Which other insects are attracted to the same substances as SLF?

Decision affected: This will also inform bait used.

What is the size and weight of the SLF at the time they usually enter the vineyards?

Decision affected: We would like to choose the optimal net size.